

DATA ANALYTICS · SQL PROJECT

Why 3 in 10 Zomato Deliveries Arrive Late

A root-cause investigation and costed recommendation, built entirely in SQL on 45,584 real delivery records.

45,584

ORDERS ANALYSED

29.8%

SLA BREACH RATE

22

CITIES

15.0%

TOP FIX'S IMPACT

EXECUTIVE SUMMARY

The Operation, At a Glance

45,584 orders measured against a 30-minute delivery promise (45 min for Semi-Urban cities)

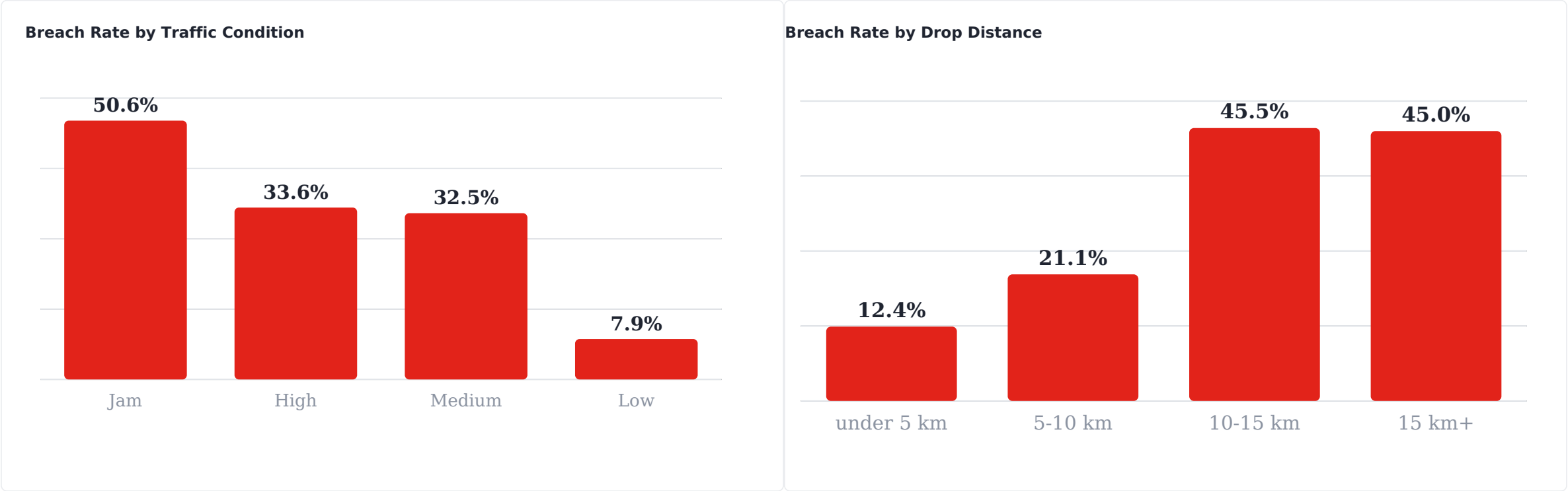


Nearly **1 in 3 orders** misses its promised delivery window. This deck walks through what actually drives that number — traffic, distance, batching, and vehicle condition — and closes with a costed ranking of which fix the business should fund first.

The headline finding, previewed: the intuitive answer — "coach the slow riders" — turns out to be the **third-best** fix available. Two structural changes (fleet maintenance, a batching cap) each outperform it.

Traffic and Distance Drive the Tail

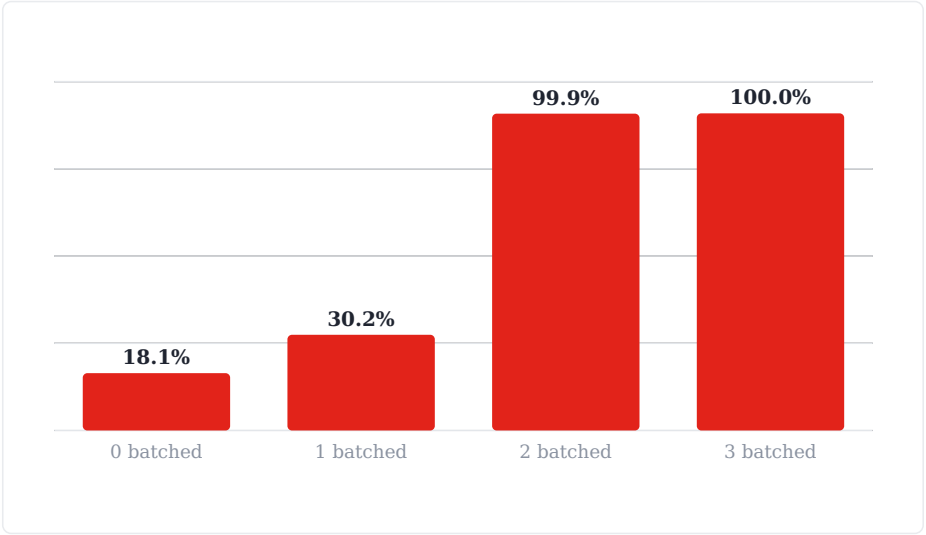
Two of the strongest single predictors of a late delivery



Jam traffic breaches 6.4× more often than Low traffic (50.6% vs 7.9%). Distance shows a threshold effect, not a slope — breach rate steps up sharply past ~10km, then flattens (45.5% → 45.0%). Past that point, a longer drop doesn't make things meaningfully worse — the 30-minute promise is simply unrealistic beyond it.

Batching Two or More Orders Is a Guaranteed Failure

A dispatch decision the business makes directly — and controls directly

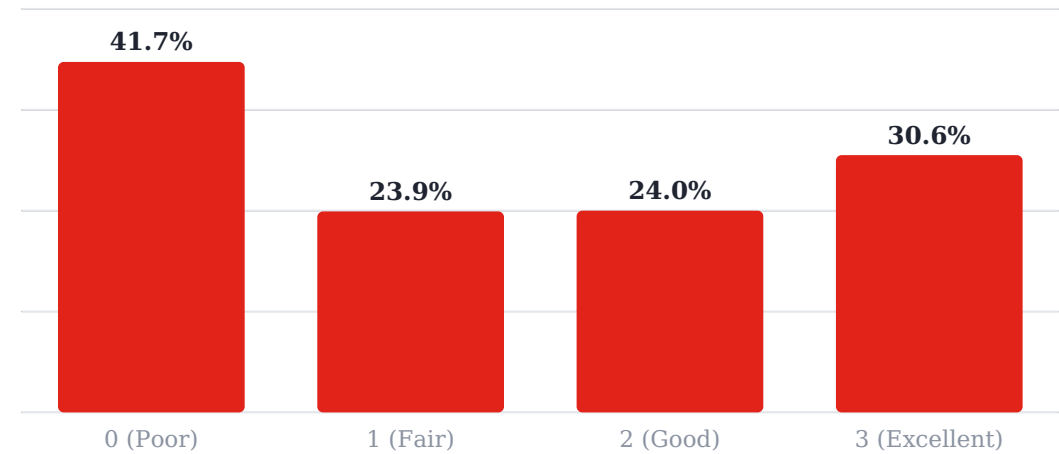


Unbatched trips breach **18.1%** of the time; a single batched order pushes that to **30.2%**. But at **two** batched orders, breach rate hits **99.9%** — and at three, **100.0%**. This is not a gradual trade-off. Under the current 30-minute promise, dispatching a third order to a rider already carrying two is a near-certainty of failure, not a risk.

Vehicle Condition Is a Cliff, Not a Gradient

The largest single controllable lever in the entire dataset

Breach Rate by Vehicle Condition (0=Poor, 3=Excellent)



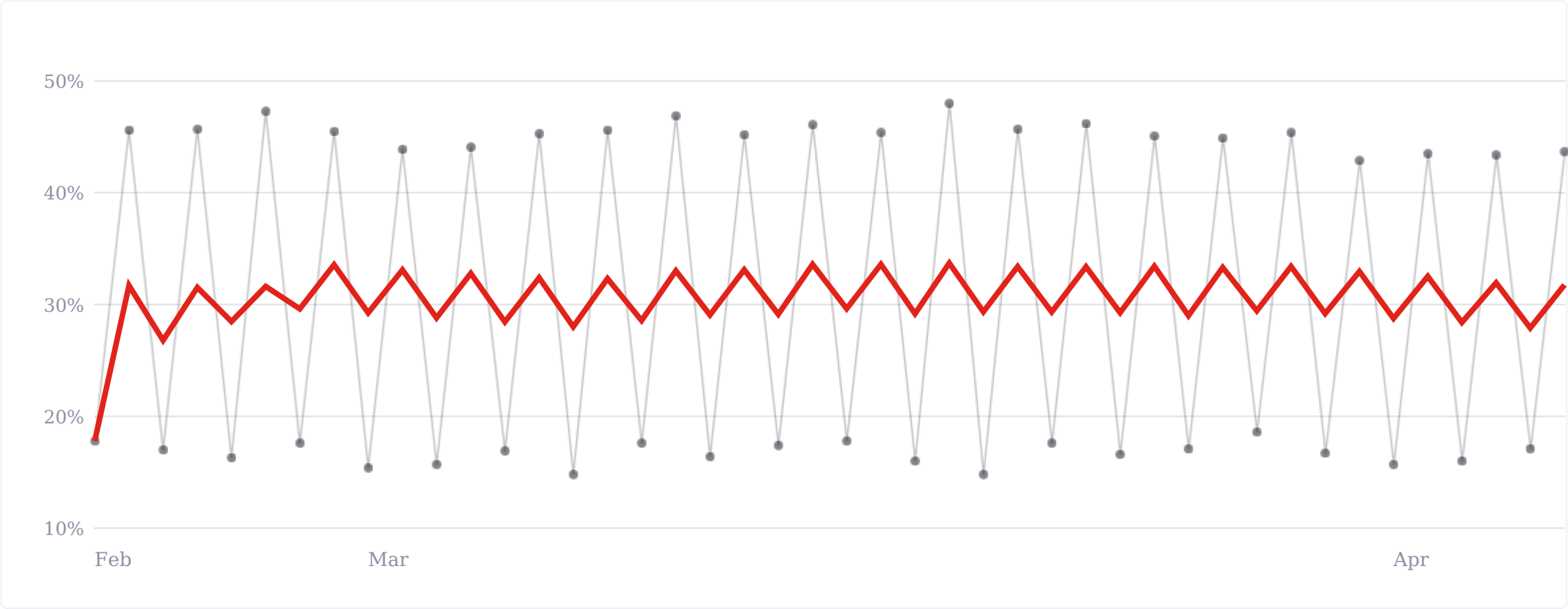
Conditions 1 and 2 are **statistically identical** (23.9% vs 24.0%). Only condition 0 — poor — is broken, at **41.7%**.

Condition 0 covers roughly **a third of the entire order book**. Fixing it is the single highest-leverage, cheapest structural change available — maintenance, not hiring.

IS THE PROBLEM GETTING WORSE?

A Stable ~30% Rate Hides a Sharp Daily Alternation

Every dot is one day's breach rate; the red line is a 7-day moving average



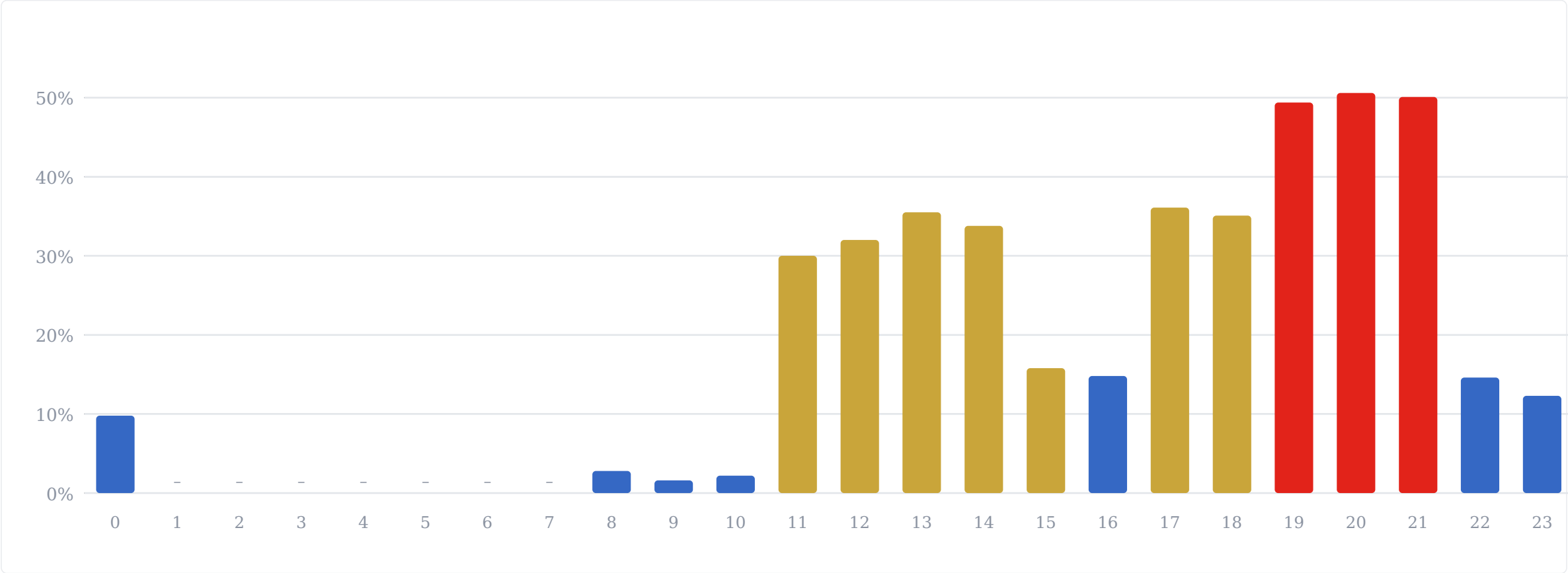
■ Daily breach rate (raw) ■ 7-day moving average

The raw daily rate alternates sharply — almost every single day swings between a ~15-19% "low" band and a ~43-48% "high" band, with no correlation to day of week. This is not a weekday/weekend pattern; it points to how the underlying orders were batched or recorded, not a real operational cycle. The 7-day average confirms the true rate is a stable 28-34%, matching the 29.8% headline figure — the problem is not worsening over the 8-week window.

TIME-OF-DAY PATTERN

The Evening Peak Is the Worst Window — But Not From Understaffing

Breach rate by hour of day. No orders were placed in hours 1–7 (overnight).



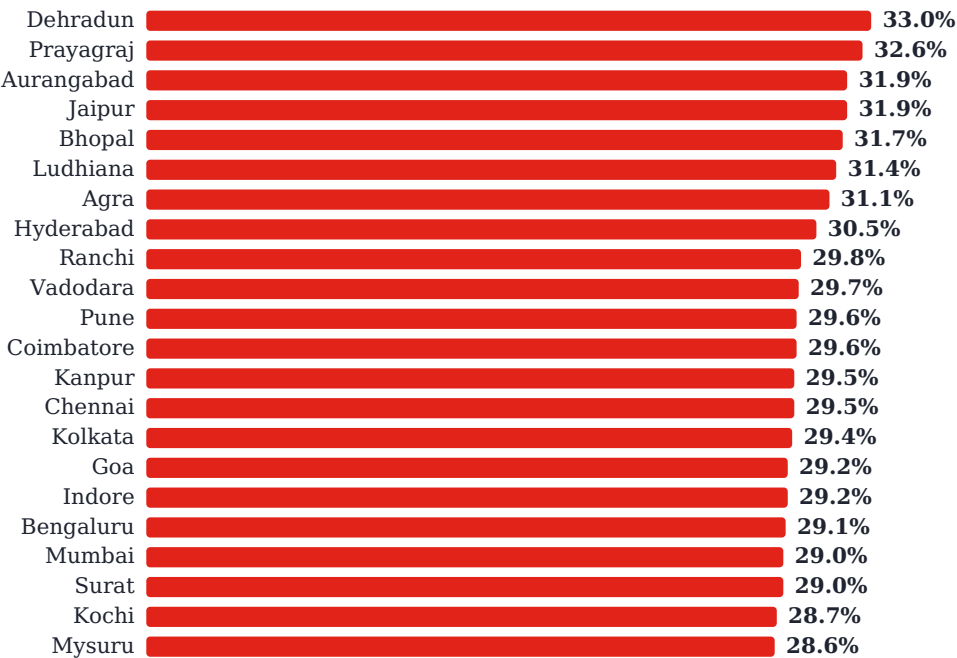
■ Under 15% (good) ■ 15-45% ■ 45%+ (worst)

Mornings (08:00–10:00) run a remarkably clean **1.6–2.8%** breach rate. The evening window **19:00–21:00** is worst at **~50%**. A smaller, easy-to-miss lunch spike (12:00–14:00, 32–35%) also stands out. Earlier analysis (Q9) found evening-peak and late-night hours carry near-identical rider load — ruling out simple understaffing as the cause of the evening spike.

No Problem City — Which Is Itself the Finding

Every one of the 22 cities lands in a narrow 4.4-point band

Breach Rate by City



28.6% – 33.0%

FULL RANGE ACROSS 22 CITIES

If a specific city were poorly managed or understaffed, its breach rate would stand out sharply from the rest. It doesn't — every city runs a near-identical rider-to-order ratio and a near-identical breach rate. This uniformity rules out "fix city X" as a strategy and points to a **systemic cause** — SLA policy and dispatch rules — not local execution.

The Best Performers, Ranked

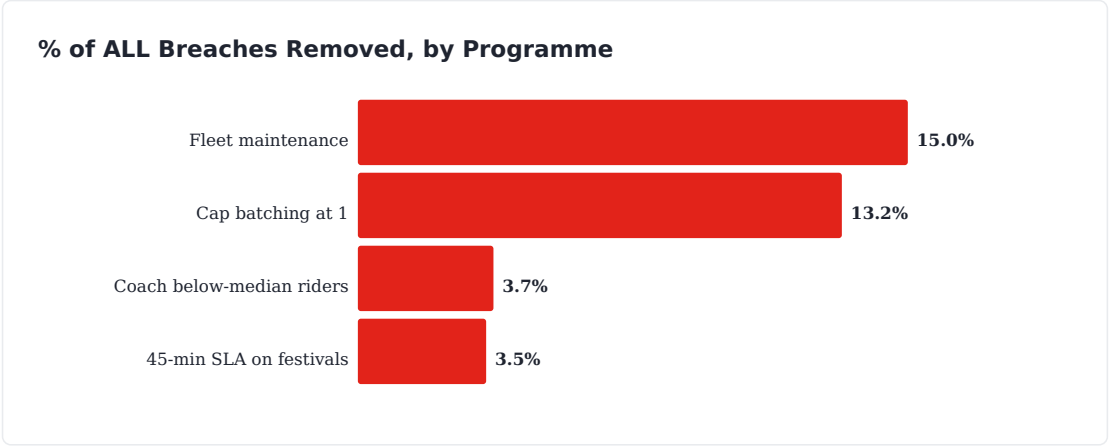
Top 10 delivery partners by lowest breach rate — a benchmark for what "good" looks like

#	PARTNER CODE	CITY	BREACH RATE
1	DEHRES20DEL03	Dehradun	0.0%
2	ALHRES11DEL01	Prayagraj	6.7%
3	KNPRES13DEL02	Kanpur	7.7%
4	KNPRES14DEL02	Kanpur	7.7%
5	AURGRES09DEL02	Aurangabad	7.7%
6	KNPRES03DEL01	Kanpur	7.7%
7	GOARES12DEL01	Goa	7.7%
8	KOCRES02DEL02	Kochi	7.7%
9	AURGRES20DEL02	Aurangabad	7.7%
10	KOLRES01DEL02	Kolkata	7.7%

These riders operate under the exact same traffic, distance, and SLA conditions as everyone else — yet post breach rates from **0.0% to 7.7%**, far below the 29.8% average. Recall from the full analysis (Q14) that a rider's star rating barely separates performance across most of the population — so this top-10 list is a genuinely useful reference point that the rating alone would not surface.

Which Fix Actually Pays — Costed and Ranked

Each programme modelled as a counterfactual: how many of the 13,597 real late orders would it have rescued?



The intuitive fix ranks third. "Coach the slow riders" — the answer most ops teams reach for first — recovers only 3.7% of all breaches. **Fleet maintenance** and a **batching cap** each outperform it, and together address roughly **4x the impact** for far less organisational effort. Note the top two programmes' scopes overlap slightly, so a combined rollout recovers somewhat less than the naive sum.

CLOSING SUMMARY

Four Things Worth Remembering

- 1 **29.8% of orders breach the SLA** — but the cause is structural (traffic, distance, batching, vehicle condition), not a worsening trend or a specific weak city.
- 2 The SLA itself breaks down past **~10km** and under **Jam traffic** — some promises are simply not achievable as currently set.
- 3 Batching **2+ orders** per trip is a near-guaranteed failure (99.9%+) — a dispatch rule the business can change tomorrow.
- 4 **Fleet maintenance** beats rider coaching as the top fix — the intuitive answer isn't always the best-funded one.